Gate Challenge I: SATLIB/CDCL Smoke Execution

First External Test of the Predeclared MAAT v1.7 Gate Protocol

MAAT Structural Selection Series – Paper 69

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Abstract

Paper 68 defined the Gate Challenge: a preregistered falsification protocol for MAAT v1.7 gate-aware structural selection. This paper reports the first external SAT/CDCL execution of that protocol on a family-balanced SATLIB smoke subset. The purpose is not to claim a full Gate Challenge result. It is to test the complete external pipeline: public benchmark staging, manifest validation, SHA256 verification, score-with-R_{rob} comparison, gate-v1.7 execution, shuffled-R_{rob} null control, and classical SAT heuristic baselines.

The SATLIB dataset gatekeeper passes: 98 manifest rows, 98 CNF files, 98 verified SHA256 hashes, 98 parseable DIMACS files, and no unexpected CNF files. A family-balanced smoke execution then uses 30 instances, five from each of six SATLIB families, with a transparent CDCL backend and conflict budget 1000.

The frozen gate does not receive support in this smoke execution. Against the fair score-based comparison, gate-v1.7 yields a positive mean utility delta but with a confidence interval crossing zero:

$$\Delta U(\text{gate} - \text{score-}R_{\mathrm{rob}}) = 7.292, \qquad 95\%\ \mathrm{CI} = [-9.493,\, 24.509].$$

Against the strongest classical baseline in this run, MOMS, the gate loses clearly:

$$\Delta U(\text{gate} - \text{MOMS}) = -66.392, \qquad 95\%\ \mathrm{CI} = [-107.679,\, -28.770].$$

The scientific value of the result is diagnostic rather than confirmatory. The SATLIB smoke test does not support the frozen gate as a practical CDCL improvement. Instead, it identifies a concrete missing channel: decision-local short-clause pressure, which is captured by MOMS but not by the current root-level robustness gate. Thus the first external SAT execution strengthens the falsifiability of the MAAT v1.7 programme while motivating a separate analysis of local structural channels in SAT.

1 Purpose and Status

Paper 69 is the first SAT/CDCL execution following the Paper-68 Gate Challenge protocol. The protocol was archived before external Gate Challenge benchmark execution:

https://doi.org/10.5281/zenodo.20882852.

This paper should be read as a family-balanced external smoke execution, not as the final SAT verdict on MAAT v1.7. Its status is intentionally narrow:

> *Does the preregistered external SAT pipeline run on public data, and what does the first family-balanced SATLIB smoke test reveal?*

The answer is mixed and useful. The data-validation and solver pipeline pass. The frozen gate does not receive positive support. The strongest classical SAT heuristic in the smoke test, MOMS, outperforms the gate, revealing a missing decision-local structural channel.

2 Preregistered Gate Protocol

The gate architecture is inherited unchanged from Paper 68. For each instance X, primary structural supports H, B, S, V define

$$R_{\text{resp}}[X] = (H[X]B[X]V[X])^{1/3}, \qquad R_{\text{rob}}[X] = \min\Big(R_{\text{resp}}[X], (H[X]B[X]S[X]V[X])^{1/4}\Big). \tag{1}$$

For SAT/CDCL, the preregistered polarity is

$$p_D = +1,$$

meaning that high robustness activates a selection/preservation mode. The calibration fold defines

$$m_D = \text{median}_{\text{cal}}(R_{\text{rob}}), \qquad s_D = \text{MAD}_{\text{cal}}(R_{\text{rob}}) + \epsilon, \qquad \epsilon = 10^{-9}. \tag{2}$$

The robustness coordinate is

$$z_R[X] = \frac{R_{\text{rob}}[X] - m_D}{s_D}. \tag{3}$$

Because the present SAT instances are static CNF files, the temporal and spatial response coordinates of the full Paper-68 gate are set to zero:

$$z_t = z_x = 0.$$

Therefore the gate used here is

$$G_{\text{gate}}[X] = \frac{1}{1 + \exp(-z_R[X])}. \tag{4}$$

This paper does not alter the gate equation, the polarity, the calibration rule, or the decision metrics after seeing the SATLIB outcomes.

3 External Dataset Staging and Validation

3.1 SATLIB source

The smoke test uses public SATLIB benchmark families. SATLIB is a long-standing public SAT benchmark archive and provides DIMACS-CNF instances across random, graph-colouring, pigeonhole, and DIMACS benchmark families. The present execution uses locally staged files only after manifest validation.

Raw SATLIB files are not redistributed as paper artifacts. The repository stores a staging script, source manifest, SHA256 hashes, and derived outputs. The local download used an explicit opt-in TLS exception because the legacy SATLIB HTTPS endpoint failed certificate verification in the local Python certificate store. This TLS mode is recorded in the manifest as audit metadata only. It does not enter any gate feature, calibration rule, solver policy, or metric.

3.2 Dataset validation gatekeeper

Before solver execution, the dataset validator checks:

- the existence of `dataset_manifest.csv`,
- required manifest columns,
- existence of every referenced CNF file,
- SHA256 equality between manifest and local file,

- DIMACS parseability,
- absence of unexpected CNF files not listed in the manifest.

The validation result is:

Check	Status
Manifest	PASS
CNFs	PASS
SHA256	PASS
Unexpected files	PASS
Dataset ready	YES

The full validation counts are shown in Table 1.

Quantity	Count
Manifest rows	98
Actual CNF files	98
Verified SHA256 files	98
Parseable DIMACS files	98
Missing manifest columns	0
Duplicate local paths	0
Missing files	0
Rows without hash	0
SHA256 mismatches	0
Parse failures	0
Unexpected CNF files	0

Table 1: Dataset validation gatekeeper for the SATLIB staging folder. This validation checks data provenance and integrity only. It does not alter the gate, calibration, or solver policy.

4 Family-Balanced Smoke Execution

The smoke execution selects five CNF files from each staged SATLIB family. This avoids the lexicographic family bias that would occur if one simply used the first N files in the folder tree. The resulting smoke subset contains 30 instances:

Family	Instances
SATLIB DIMACS AIM	5
SATLIB DIMACS Dubois	5
SATLIB DIMACS pigeonhole	5
SATLIB graph-colouring flat30	5
SATLIB random 3-SAT uf20	5
SATLIB random 3-SAT uuf50	5

The stable hash split assigns 6 instances to calibration and 24 instances to test. The transparent CDCL backend is inherited from the Paper-63 infrastructure and is used here for auditability, not as a state-of-the-art SAT solver. The conflict budget is 1000 for this smoke execution.

The policies compared are:

- VSIDS,

- MOMS,
- Jeroslow–Wang,
- progress-only,
- score-with-R_{rob},
- gate-v1.7,
- shuffled-R_{rob} gate null.

The compute cost is the predeclared CDCL utility proxy:

$$C = N_{\mathrm{conflict}} 0.30\, N_{\mathrm{decision}} 0.004\, N_{\mathrm{propagation}} N_{\mathrm{timeout}}\, B_{\mathrm{conflict}},$$

where $B_{\mathrm{conflict}} = 1000$ in the present smoke execution.

5 Results

5.1 Gate versus score-with-R_{rob}

The fair internal Gate Challenge comparison is gate-v1.7 versus score-with- R_{rob}. The smoke result is shown in Table 2. Positive ΔU means lower compute cost for the gate.

Comparison	Pairs	Mean ΔU	95% CI low	95% CI high	Wins/Losses/Ties
Gate vs score-with-R_{rob}	24	7.292	-9.493	24.509	11/13/0
Gate vs shuffled-R_{rob}	24	2.659	-7.627	13.909	13/10/1

Table 2: Fair gate comparison and shuffled-R_{rob} null comparison. The gate has positive mean utility delta but the confidence interval crosses zero. This is not support under the Paper-68 criterion.

The result is therefore:

> *The SATLIB smoke test does not support the frozen gate as an improvement over score-with-R_{rob}.*

5.2 Gate versus classical SAT heuristics

The classical baseline comparison is more severe. Table 3 shows that MOMS is the strongest baseline in this smoke test. The gate loses to MOMS with a confidence interval entirely below zero.

Comparison	Pairs	Mean ΔU	95% CI low	95% CI high	Wins/Losses/Ties
Gate vs VSIDS	24	-23.758	-98.753	17.718	12/12/0
Gate vs MOMS	24	-66.392	-107.679	-28.770	3/20/1
Gate vs Jeroslow–Wang	24	-9.439	-49.772	21.056	7/16/1
Gate vs progress-only	24	-33.495	-103.599	4.715	8/16/0

Table 3: Gate-v1.7 compared against classical and progress baselines. The MOMS comparison is the most important negative smoke-test result.

Figure 1 shows policy-level compute costs. Figure 2 shows paired utility deltas with bootstrap confidence intervals.

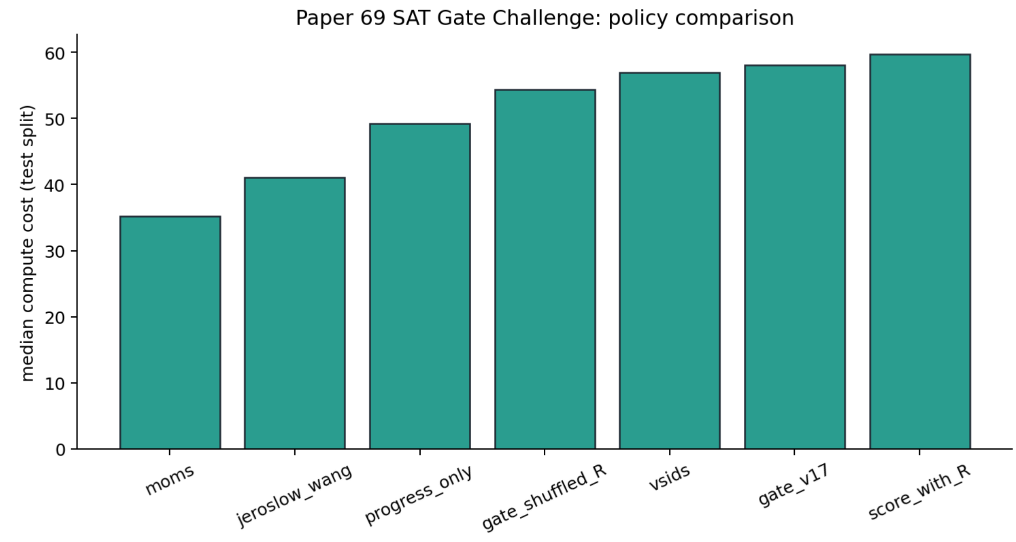

Figure 1: Policy compute-cost summary for the family-balanced SATLIB smoke execution. The figure is diagnostic only and should not be read as a full Gate Challenge result.

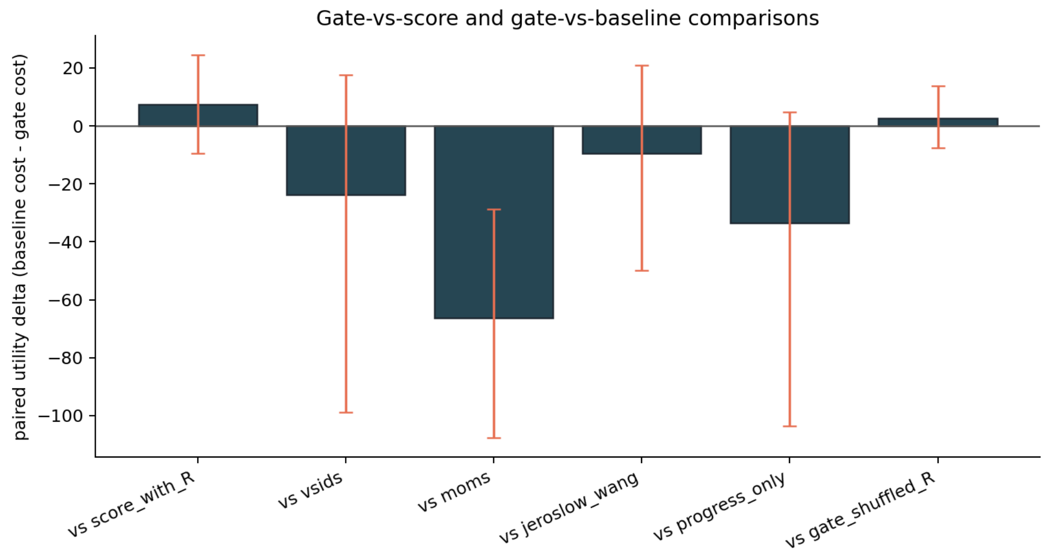

Figure 2: Paired utility comparisons. Positive values favour the gate. The gate does not achieve a positive lower confidence bound against score-with-R_{rob} and loses clearly to MOMS in this smoke test.

6 Why MOMS Wins

The most useful result of this smoke execution is not that the gate loses. It is that the loss is structurally interpretable.

MOMS is a decision-local short-clause heuristic. It prioritizes variables appearing in clauses of minimum current length. In contrast, the frozen v1.7 gate used here is a root-level robustness activation computed from instance-level supports. These are different information channels.

The post-hoc diagnostic asks:

> *Which structural information is used by MOMS that the frozen v1.7 gate does not use?*

The family-level regret analysis identifies the largest mean gate-to-MOMS regret in:

Dubois, pigeonhole, AIM.

These are precisely families where local constraint pressure and short-clause structure are expected to matter.

Family	Mean gate cost	Mean MOMS cost	Mean regret to MOMS
SATLIB DIMACS AIM	65.974	37.938	28.036
SATLIB DIMACS Dubois	263.394	65.386	198.008
SATLIB DIMACS pigeonhole	2185.692	2097.555	88.137
SATLIB graph-colouring flat30	5.432	5.203	0.229
SATLIB random 3-SAT uf20	3.993	3.022	0.971
SATLIB random 3-SAT uuf50	59.702	40.646	19.057

Table 4: Family-level MOMS diagnostic. The strongest MOMS advantage appears in Dubois and pigeonhole instances. This suggests that local short-clause pressure carries information not captured by the current root-level gate.

Figures 3 and 4 visualize the same diagnostic.

Figure 3: Family-level regret of gate-v1.7 relative to MOMS. Large positive values indicate families where MOMS uses decision-local information that the frozen gate does not capture.

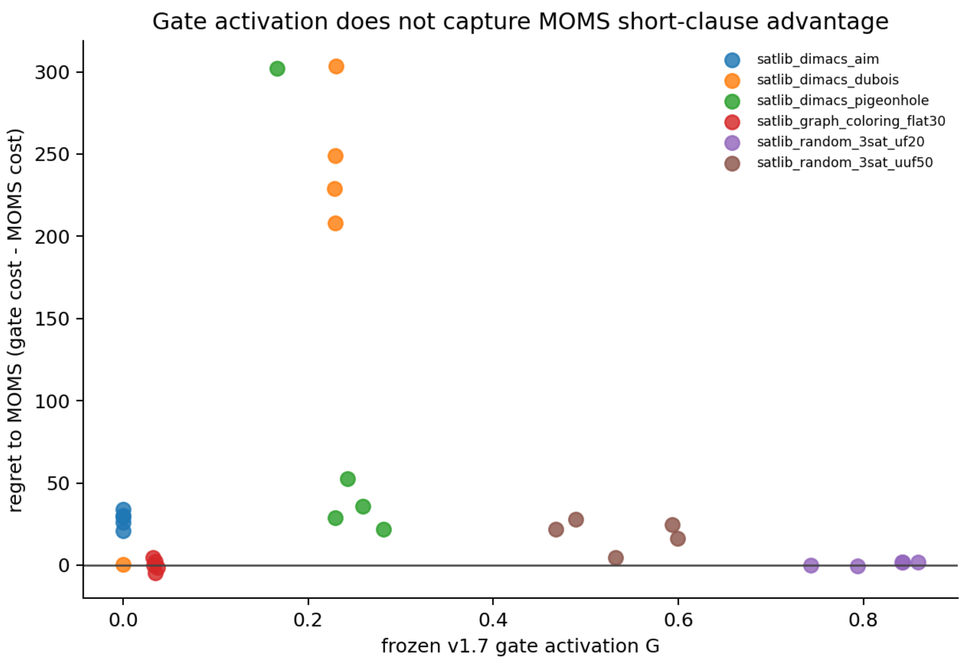

Figure 4: Gate activation versus regret to MOMS. The absence of a clear monotonic relation reinforces the interpretation that root-level robustness activation is not measuring the same channel as decision-local short-clause pressure.

7 Interpretation

The key result is negative but informative:

> *The SATLIB smoke test does not support the frozen gate as a practical CDCL improvement. Instead, it identifies a concrete missing channel: decision-local short-clause pressure, captured by MOMS but not by the current root-level robustness gate.*

This should not be interpreted as a reason to tune the gate after the fact. The Paper-68 protocol exists precisely to prevent that. Changing thresholds, redefining the gate, or adding short-clause information to the gate after seeing the SATLIB smoke result would invalidate the present test.

Instead, the appropriate scientific conclusion is:

> *For SAT/CDCL, local structural channels appear more important than the current global robustness gate.*

This shifts the next question from "How can MAAT beat MOMS?" to "Which local structural channel does MOMS exploit, and can it be described independently of MOMS itself?"

8 Limitations

The limitations are substantial:

- The execution is a smoke test, not a full Gate Challenge result.
- The subset contains 30 SATLIB instances, five per family.
- The transparent Python CDCL backend is auditable but not state-of-the-art.

- LRB, CHB, CaDiCaL, Glucose, Kissat, and MiniSat are not yet included.
- The conflict budget is 1000.
- The family-balanced selection is deterministic but small.
- The MOMS analysis is post-hoc diagnostic, not a preregistered new gate.

These limitations do not undermine the smoke-test conclusion. They define the boundary of the claim. The present paper reports that the first external SAT smoke execution passes data validation and pipeline reproducibility, but does not support the frozen gate in SAT/CDCL.

9 Reproducibility

The experiment folder is:

`experiments/gate_challenge_sat_paper69/`.

Dataset validation:

```
python3 gate_challenge_sat_paper69.py --validate-only
```

Family-balanced smoke execution:

```
python3 gate_challenge_sat_paper69.py --family-balanced 5 --conflict-budget 1000
```

Post-hoc MOMS diagnostic:

```
python3 analyze_moms_vs_gate_paper69.py
```

Primary outputs:

- `paper69_dataset_validation.json`,
- `paper69_summary.json`,
- `paper69_solve_records.csv`,
- `paper69_gate_comparisons.csv`,
- `paper69_moms_vs_gate_diagnostics.json`,
- `paper69_moms_vs_gate_family_diagnostics.csv`.

The repository is:

https://github.com/Chris4081/structural-selection-principle.

10 Data Attribution and License Note

SATLIB benchmark files are external public benchmark data. They should be cited to the original SATLIB benchmark archive and associated authors or maintainers when reused. The present repository does not intentionally redistribute raw SATLIB CNF files as primary paper artifacts. It provides staging scripts, SHA256 manifests, derived CSV/JSON outputs, and figures. No endorsement by SATLIB maintainers, SAT Competition organizers, DIMACS benchmark authors, or any external dataset provider is implied.

11 Conclusion

Paper 69 executes the first external SAT/CDCL smoke test of the preregistered MAAT v1.7 Gate Challenge protocol. The data-validation gatekeeper passes on 98 staged SATLIB CNFs. A family-balanced 30-instance SATLIB smoke execution then tests the frozen gate against score-with-R_{rob}, shuffled-R_{rob}, and classical CDCL heuristics.

The result does not support the frozen gate. The gate has a positive mean utility delta against score-with-R_{rob}, but the confidence interval crosses zero. Against MOMS, the gate loses clearly.

The important scientific output is therefore not confirmation. It is localization of the missing channel. MOMS appears to exploit decision-local short-clause pressure that the current root-level robustness gate does not encode.

Thus Paper 69 strengthens the MAAT programme by reporting an external negative smoke result under a preregistered protocol. It also motivates the next analysis question:

> *Which local structural channels in SAT explain the success of classical short-clause heuristics, and when do global structural gates fail?*

References

[1] C. Krieg, "The Gate Challenge: A Predeclared Falsification Protocol for MAAT v1.7 Gate-Aware Structural Selection," MAAT Structural Selection Series, Paper 68 (2026). Preregistration archive: https://doi.org/10.5281/zenodo.20882852.

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[4] R. G. Jeroslow and J. Wang, Solving propositional satisfiability problems, *Annals of Mathematics and Artificial Intelligence* **1**, 167–187 (1990).

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